

Handout 02.11: Interest Math Practice

Name: _____ Date: _____ Partner: _____

Rates on this sheet are typical for 2026 and marked. The people are made up.

My do now guess: a \$500 phone at \$25 a month on a store card will cost \$_____ in total.

Part A: Debit vs credit, side by side

	Debit card	Credit card
Whose money is it?		
When does it leave?		
Is there a bill?		
Is there interest?		
What if the card is stolen?		
Does it build a credit score?		

Credit is _____

Debt is _____

Interest is _____

Part B: Simple interest on \$500

The formula: $I = P \times r \times t$ (interest = amount borrowed x rate as a decimal x years)

Rates as decimals: 5 percent = 0.05. 15 percent = 0.15. 24 percent = 0.24.

Worked example: \$500 at 5 percent for 1 year. $\$500 \times 0.05 \times 1 = \25 . Total owed: $\$500 + \$25 = \$525$.

Rate	What it is like	One year: interest	One year: total owed	Three years: interest	Three years: total owed
5 percent	a good car loan	$\$500 \times 0.05$ $\times 1 = \$$ _____	$\$$ _____	$\$500 \times 0.05$ $\times 3 = \$$ _____	$\$$ _____
15 percent	a credit union card	$\$500 \times 0.15$ $\times 1 = \$$ _____	$\$$ _____	$\$500 \times 0.15$ $\times 3 = \$$ _____	$\$$ _____
24 percent	a typical bank credit card	$\$500 \times 0.24$ $\times 1 = \$$ _____	$\$$ _____	$\$500 \times 0.24$ $\times 3 = \$$ _____	$\$$ _____

At which rate does three years of interest cost more than half the phone? _____

(Grade 6: one-year columns only.)

Part C: Two more loans

1. Ines borrows \$1,200 on a store card at 15 percent and pays nothing for two years.
Interest: \$_____ x _____ x _____ = \$_____. Total owed: \$_____.
2. A used car loan: \$8,000 at 5 percent for four years. Interest: \$_____ x _____ x _____ = \$_____. Total paid: \$_____.
3. Which loan cost more in dollars? _____ Which cost more as a percent of what was borrowed? _____ How can both be true?

Part D: The minimum payment trap

Leo puts a \$500 phone on a credit card at 24 percent APR. That is 2 percent a month. Each month the card adds 2 percent of the balance, then his payment comes off.

The first three months at \$25 a month (follow along on the board):

Month	Balance at start	Add 2 percent	Pay \$25	Balance at end
1	\$500.00	\$10.00	\$25.00	\$485.00
2	\$485.00	\$9.70	\$25.00	\$469.70
3	\$469.70	\$9.39	\$25.00	\$454.09

Three months in, Leo has paid \$75 and still owes \$454.09. Copy the finished results from the slide:

Plan	Months to pay off	Total interest	Total paid for the \$500 phone
\$25 a month (the minimum)			
\$50 a month			
Paid in full the first month			

My recommendation (two sentences, with the numbers):



My do now guess was \$ _____. The real number was \$ _____. I was off by \$ _____.

Part E: The credit score, in one paragraph

A credit score is a number from _____ to _____ that shows _____.

It goes up when I _____ and _____.

It goes down when I _____ or _____.

It matters because _____, _____, and sometimes _____ look at it.

Grade 8 stretch

1. Do months 4 and 5 of the \$25 plan by hand and check them against the pattern.
2. What monthly payment would clear the \$500 in ten months? (Try \$55, then \$56.)
About how much interest is that?
3. Three cards: a bank card at 24 percent, a store card at 29 percent, a credit union card at 15 percent. A person with a good score should pick _____ because _____.

Exit card (3-2-1)

3 differences between credit and debit: _____, _____, _____

2 things that move a credit score: _____, _____

1 number: the total cost of the \$500 phone at \$25 a month: \$ _____